🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.6

Explore Integers and Their Opposites

Lesson 7-1 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use integers to represent quantities in everyday life, explain what 0 means in each situation, and name the opposite of an integer on the number line.

Language Objective: I can explain my reasoning using the words integer, opposite, negative, positive, and number line.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Integer | Opposite | Negative number | Positive number | Number line

👆 *Tap any word to see what it means and a picture.*

1. A counting number, its opposite, or 0 is an __________.

2. A number the same distance from 0 but on the other side is the __________.

3. A number less than 0, written with a – sign, is a __________.

4. A number greater than 0 is a __________.

5. A line with numbers placed in order is a __________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Write each play as an integer, then decide: did the team make a first down, and how far are they from the line of scrimmage?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Integer** — The counting numbers, their opposites, and 0.
  Número entero — Los números para contar, sus opuestos y el 0.

- [ ] **Opposite** — A number that is the same distance from 0 as another number, but on the other side of 0.
  Opuesto — Un número que está a la misma distancia de 0 que otro número, pero del otro lado del 0.

- [ ] **Negative number** — A number less than 0, written with a – sign. On a number line it sits to the left of 0.
  Número negativo — Un número menor que 0, escrito con un signo –. En la recta numérica queda a la izquierda del 0.

- [ ] **Positive number** — A number greater than 0. On a number line it sits to the right of 0.
  Número positivo — Un número mayor que 0. En la recta numérica queda a la derecha del 0.

- [ ] **Number line** — A straight line where numbers are placed in order. It extends to the left of 0 to show values less than 0.
  Recta numérica — Una recta donde los números se colocan en orden. Se extiende a la izquierda del 0 para mostrar valores menores que 0.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The 10-yard loss is the integer ____ and the 3-yard loss is the integer ____. Combining all four plays leaves the team ____ yards from where they started, which is ____ the line of scrimmage. Because they needed at least 10 yards, the team ____ make a first down.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: In this situation 0 means ____ because ____.

Evidence: Student explains 0 is a chosen reference point (sea level), so values above it are positive and values below it are negative.

Reasoning: This evidence connects the definition of integer to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The 10-yard loss is the integer ____ and the 3-yard loss is the integer ____. Combining all four plays leaves the team ____ yards from where they started, which is ____ the line of scrimmage. Because they needed at least 10 yards, the team ____ make a first down.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Integer
2. **Notes 2:** Opposite
3. **Notes 3:** Negative number
4. **Notes 4:** Positive number
5. **Notes 5:** Number line
6. **Watch for this mistake:** *The most common mistake in this lesson is thinking that the opposite of a negative integer is still negative — for example, saying the opposite of -7 is -7, or that the opposite of -12 is -24. Opposites are the same DISTANCE from 0 but on OPPOSITE sides of 0, so taking an opposite always changes the direction and never changes the distance: the opposite of -7 is 7 and the opposite of -12 is 12. The one exception is 0, which is 0 units from itself and is therefore its own opposite. Before you submit, count the units from 0 and check that your answer is on the other side.*
7. **Try It 1:** A. 12°C and -12°C — *The opposite of -12 is 12, so 2 p.m. was 12°C. The opposite of 12 is -12, so 11 p.m. was back to -12°C.*
8. **Try It 2:** A. 12 yards past the line of scrimmage — *Write each play as an integer: -10, 4, -3, 21. Combining them gives -10 + 4 - 3 + 21 = 12, so the team ends 12 yards past the line of scrimmage.*